

Case Report



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Conflict of Interest

The author declares that they have no competing interests.

Nutritional Management in Severe Burn Patients: A Case Report

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ABSTRACT

Patients with severe burns frequently experience inadequate nutrition due to hypermetabolism and its associated complications, substantially increasing the risk of malnutrition. This case report describes the nutritional intervention for a 54-year-old male patient admitted with total body surface area burns of 42.4%, including 15% third-degree burns caused by flames. It highlights the importance of active nutritional support and continuous monitoring during the management of complex burn cases. Upon admission, the patient's nutritional intake was restricted due to fluid resuscitation, frequent surgeries requiring fasting, renal dysfunction, and gastrointestinal complications. Nutritional requirements were calculated using the Harris-Benedict and Toronto equations; however, it was difficult to meet the targeted nutritional demands during the initial Nutrition Support Team (NST) consultation due to renal dysfunction and hemodynamic instability. Subsequent efforts, including oral nutritional supplements and adjunctive parenteral nutrition, were implemented; however, multifactorial issues, such as systemic deterioration and complications, further exacerbated the patient's nutritional status. As a result, the patient experienced a 15% reduction in his usual body weight, decreasing from 100 kg to 85 kg. This case underscores the vital role of proactive NST involvement and ongoing nutritional intervention in the management of patients with severe burns and complex complications.

Keywords: Burns; Metabolism; Malnutrition; Nutritional support

INTRODUCTION

Severe burns are among the most fatal types of trauma, with associated risks significantly increasing if tailored nutritional support is not provided [1]. Burn injuries induce hypermetabolism and multiple complications, including whole-body edema, shock, acute kidney injury, and systemic inflammatory response syndrome (SIRS). These injuries increase the demand for anabolic processes in protein and energy metabolism, while simultaneously triggering a systemic catabolic state. This metabolic imbalance leads to malnutrition, accelerated muscle wasting, and immune suppression, increasing the risk of infections and related complications. Therefore, addressing these metabolic demands through appropriate nutritional interventions is crucial for improving the clinical outcomes of patients with

burns [2]. Although fluid resuscitation is essential for treating burn shock in patients with severe burns during the early stages of treatment [3], it may limit the provision of adequate nutritional support. As patients' conditions fluctuate, frequent variations in nutritional supply and methods arise, complicating adequate nutritional delivery. Consequently, patients are at high risk of malnutrition, which exacerbates the risk of infectious complications and increases mortality.

Patients with severe burns are known to have metabolic rates that are 120%–150% higher than normal [4]. For burns involving more than 40% of the total body surface area (TBSA), energy expenditure can continue to increase for 1–4 weeks post-injury [2]. The current study recommends using indirect calorimetry to determine the energy requirements of patients with burns. In the absence of indirect calorimetry, the American Society for Parenteral and Enteral Nutrition (ASPEN) guidelines recommend using the Harris-Benedict equation with a 1.5 stress factor, Toronto equation, or Ireton-Jones equation [5], whereas the European Society for Clinical Nutrition and Metabolism (ESPEN) guidelines suggest using Toronto equation [3]. However, these predictive equations lack a strong correlation with the measured energy expenditure in burn patients [6]. Therefore, energy requirements must be individualized and their adequacy must be regularly evaluated. Protein intake of 1.5–2 g/kg/day is recommended [3], constituting 15%–25% of the total energy requirement [7]. Obese patients with a body mass index (BMI) of 30–40 should receive 2.0 g/kg/day of protein based on their ideal body weight, while those with a BMI exceeding 40 should receive 2.5 g/kg/day [8].

Burn wounds lead to a significant losses of micronutrients and body fluids through the skin, making it challenging to maintain adequate nutrient levels [8]. Therefore, vitamin and mineral supplements are critical. However, the use of glutamine supplementation remains controversial and requires caution. Although glutamine correlates with reduced mortality in traumatic injury cases, recent studies on patients with severe burns questions its efficacy and safety [2]. Regular monitoring of blood test results and ongoing evaluation of nutrient intake adequacy are essential. Nutritional interventions play a positive role in wound healing and improve treatment outcomes in patients with severe burns.

In conclusion, this case report discusses the nutritional intervention process and outcomes in a patient with severe burns admitted to our hospital, highlighting the critical role of the Nutrition Support Team (NST) and the need for a multidisciplinary approach in managing severe burns. This study was approved by the Institutional Review Board of the Hanil General Hospital (Approval No. HGH 2024-11-005).

CASE

A 54-year-old male patient with a history of diabetic neuropathy, diabetic retinopathy, and suspected diabetic chronic kidney disease sustained flame burns covering 42.4% of the TBSA (15% third-degree burns). At admission, his height was 178 cm, and weight was 108.5 kg (usual weight: 100 kg). The changes in patient weight and blood test results during hospitalization are summarized in **Table 1**.

During hospitalization, the patient underwent multiple surgeries and fasting, experienced cardiac and renal dysfunction, and faced challenges in receiving adequate nutrition. Consequently, his weight decreased from 100 kg to 85 kg, reflecting muscle loss and

Table 1. Changes in weight and blood test values

Variables	Normal range	HD #1	HD #2	HD #3	HD #7	HD #10	HD #16	HD #28	HD #35	HD #38	HD #42	HD #49	HD #65	HD #70	HD #133
Body weight (kg)		108.5	116	114.5	120.0	123.5	113.5	107.0	111.0	106.0	103.0	101.8	105.0	103.6	83.7
Laboratory data															
Albumin (g/dL)	3.5–5.2	2.4	1.7	2.4	2.3	2.0	2.9	2.2	2.4	2.0	2.2	2.2	2.3	2.3	-
Prealbumin (mg/dL)	20–40	-	-	-	5.27	-	-	10.45	10.04	-	-	-	-	12.19	-
Blood urea nitrogen (mg/dL)	7–20	42	55	47	78	74	49	33	24	18	22	26	43	15	19
Creatinine (mg/dL)	0.67–1.17	2.68	2.81	2.40	3.48	3.38	2.81	2.35	2.03	2.07	1.59	1.40	1.83	1.37	1.51
Sodium (mmol/L)	136–146	132	132	133	124	134	143	134	134	137	138	134	134	128	135
Potassium (mmol/L)	3.5–5.1	5.4	4.9	4.8	4.3	3.8	4.1	3.7	3.6	3.4	3.9	4.4	4.3	3.7	4.7
Hemoglobin (g/dL)	13–17	16.8	12.8	8.8	9.9	10.1	12.1	8.3	11.7	9.3	8.7	11.0	9.3	9.0	9.4
Magnesium (mg/dL)	1.9–2.5	1.6	-	1.5	4	2.2	-	1.2	-	1.3	-	-	-	-	-
Calcium (mmol/L)	1.13–1.32	-	-	0.98	1.04	0.98	1.09	0.95	0.94	1.03	1.10	1.06	1.12	-	-
Zinc (ug/dL)	66–110	-	-	-	49	-	-	65	95	-	-	-	-	87	-
Manganese (ug/L)	0.4–6.2	-	-	-	-	-	-	1.26	1.14	-	-	-	-	0.66	-
Copper (ug/dL)	68–140	-	-	-	96	-	-	112	90	-	-	-	-	53	-
Selenium (ug/dL)	95–165	-	-	-	77	-	-	75	65	-	-	-	-	61	-
Chromium (ug/dL)	0–3	-	-	-	-	-	-	-	2.43	-	-	-	-	1.03	-
C-reactive protein (mg/dL)	0–5	11.1	147.0	105.8	65.0	47.0	80.3	12.6	25.4	15.4	3.5	-	-	-	-
Glucose (mg/dL)	70–100	389	226	270	354	265	157	83	66	112	99	76	104	133	166
AST (IU/L)	5–35	27	26	21	39	86	67	82	59	66	124	43	29	18	34
ALT (IU/L)	5–45	19	15	12	28	92	63	91	32	29	80	20	53	22	26
Myoglobin (ng/mL)	17.4–105.7	470.8	-	-	-	-	-	-	-	-	-	-	-	-	-

HD, hospital day; AST, aspartate aminotransferase; ALT, alanine aminotransferase.

malnutrition. The NST was consulted on hospital day (HD) #3 and again on HD #44 for adequate nutritional management. **Table 2** provides a summary of the NST interventions.

On the first day of admission, the patient was administered over 10 liters of intravenous fluids, including dextrose, albumin, amino acids, and micronutrient supplements. On the second day, oral feeding was initiated. however, fasting was resumed the following day

Table 2. Summary of nutrition interventions from the Nutrition Support Team

Variables	HD #3	HD #44
Patient status	• Surgery day • Body weight: 114.5 kg • NPO • Ventilation care • Mentally confused	• Body weight: 104.5 kg • Mentally alert
Nutritional requirements	• Energy: 3,300 kcal • Protein: 200 g	• Energy: 3,000 kcal • Protein: 230 g
Nutrition intake	• Winuf central 1,435 mL Inj. • Dipeptiven 100 mL Inj. • TamiPool Inj. • Multiblue 5 10 mL Inj. • Zinc-in 20 mL Inj. • Calcium gluconate 2 g/20 mL Inj. • Vitamin K1 10 mg/1 mL Inj. • 5% dextrose 1,000 mL Inj. • Total energy: 1,820 kcal; total protein: 92.9 g	• Consumed 50% or less of a high-protein diet • Encover solution 200 mL × 4 • Total energy: 2,200 kcal; total protein: 80 g
Recommendation	• If TPN is being supplied without any issues, increase the TPN dose while monitoring renal function • Since achieving adequate intake is challenging in the early dietary phase, TPN should be continued in parallel	• Encourage a high-protein diet and Encover intake (6 packs per day). • Add Proamine 10% (500 mL intravenous infusion)

HD, hospital day; NPO, nothing by mouth; TPN, total parenteral nutrition; Inj., injection.

because of escharectomy, prompting the initiation of total parenteral nutrition (TPN) and NST intervention.

HD #3: first NST consultation

Given the absence of indirect calorimetry in our facility, energy requirements were estimated using the ASPEN and ESPEN recommended formulas. The modified Harris-Benedict equation was used, which adjusted for the stress factor associated with the extent of burns [2]. The Toronto equation was used to set nutritional targets. Energy requirements were estimated at 3,300 kcal and protein at 200 g. At that time, the patient was receiving TPN and intravenous amino acid solutions, which met only 55% of estimated energy needs and 45% of protein requirements. Severe hypovolemia and renal dysfunction, following the burn, limited the ability to increase nutritional support. However, with no progression to renal failure, gradual increases in nutritional supply were implemented, while monitoring the patient's nutritional status. To avoid overfeeding associated with the Harris-Benedict equation, the Toronto equation was used to establish an initial target of approximately 70% of the Harris-Benedict estimate, with 25% of the total energy provided as protein [7]. Vitamin and mineral supplementation continued during TPN administration. The team recommended transitioning to oral feeding as soon as the patient's condition stabilized.

Oral feeding resumed but was inadequate due to nausea and dyspepsia. Intravenous amino acid supplementation continued, and caloric intake was restricted to less than 3,000 kcal due to challenges in controlling blood glucose levels. On HD #8, the patient developed dyspnea while eating, requiring endotracheal intubation with mechanical ventilation. Paralytic ileus necessitated nasogastric drainage, and the patient developed septic shock, leading to prolonged fasting. Worsening renal function during septic shock required fluid restrictions, further limiting parenteral nutrition administration.

On HD #12, surgery was interrupted because of increased intraoperative bleeding and hypothermia. Severe pulmonary edema and fluid retention from worsening renal function necessitated hemodialysis. On HD #22, oral feeding was resumed; however, the patient exhibited mental confusion and insufficient intake, requiring additional intravenous amino acid supplementation. Persistent constipation lasting over 12 days was managed with laxatives. Following subsequent surgery, the patient's general condition improved, allowing transfer to the general ward on HD #30. However, on HD #32, desaturation after surgery led to reintubation and intensive care unit readmission.

Subsequently, the patient's condition improved, and oral feeding was restarted; however, intake remained inadequate, prompting the addition of oral nutritional supplements (ONS). Persistent diarrhea further hindered adequate nutritional intake. On HD #39, the patient developed atelectasis and underwent a fourth hemodialysis session due to fluid retention following surgery. Pleural effusion and desaturation necessitated reintubation, leading to the initiation of enteral nutrition via tube feeding, supplemented with TPN to meet nutritional requirements.

HD #44: second NST consultation

The patient's nutritional status had deteriorated, with several additional planned surgeries and significant exudate loss from the burn area. An NST was consulted to improve the patient's overall condition and facilitate surgery. The estimated nutritional requirements, accounting for burn-related losses and surgical needs, were 3,000 kcal and 230 g of protein.

Intake from hospital meals and ONS met approximately 75% of energy needs and 35% of protein requirements. The NST recommended adding intravenous amino acid solutions.

Two days later, food intake worsened due to nausea, dyspepsia, and diarrhea, complicating efforts to increase nutritional intake. To prepare for upcoming surgeries, intermittent provision of à la carte meals is recommended to encourage oral intake. Although hemodialysis was not required at this stage, monitoring of the electrolyte levels and renal function was advised. Fluid intake was restricted to prevent fluid overload, while solid food intake was encouraged.

Despite these efforts, oral intake remained inadequate; worsening renal function required resuming hemodialysis, and poor nutritional status frequently delayed planned surgeries. After transitioning to a renal diet, therapeutic dietary management was tailored to the patient's preferences and eating abilities to encourage food intake. On HD #134, 99% of the burn wounds had healed, and the patient was discharged on HD #200 after completing orthopedic treatments.

DISCUSSION

Severe burns increase energy and protein requirements due to hypermetabolism and protein loss at wound sites [9]. During the first 1–2 days post-injury, substantial volumes of fluids are administered for resuscitation [4]. While patients with burns have significant nutritional demands, the extensive fluid requirements for resuscitation pose challenges to providing adequate parenteral nutrition. Furthermore, systemic metabolic alterations, such as burn shock and vasopressor use, often hinder enteral nutrition. These challenges reduce the patient's immunity and treatment efficacy, increasing the risk of malnutrition. In this case, the patient sustained burns covering 42.4% of the TBSA and, from the early stages of hospitalization, exhibited inadequate nutritional intake due to renal dysfunction, repeated fasting for surgical procedures, and gastrointestinal symptoms. This led to significant weight loss and nutritional decline.

Based on these observations, we discuss the underlying causes of these outcomes and propose strategies to improve them.

Initially, fluid resuscitation and renal impairment, compounded by multiple surgeries, fasting, facial burns, and gastrointestinal symptoms, restricted nutritional intake—particularly oral intake—exacerbating energy deficit and weight loss. Although the primary objective of nutritional therapy for patients with burns is to prevent weight loss exceeding 10% of usual body weight [10], this patient experienced a 15% reduction, from 100 kg to 85 kg.

Substantial amounts of protein are required to compensate for losses at the wound site and support recovery. However, renal impairment restricts the patient's ability to increase protein intake sufficiently to meet metabolic demands. Despite intermittent hemodialysis when indicated, fluid restrictions imposed to manage volume overload hindered adequate nutritional support, thereby delaying wound healing and prolonging hospitalization.

The dynamic and rapidly changing conditions of patients with burns necessitate continuous monitoring, as their metabolic and nutritional demands fluctuate over time and require adaptive nutritional support. Calculated energy requirements, derived from

predictive equations, provide a foundational reference; however, they should not be treated as absolute. Ongoing nutritional status assessment is crucial for adjusting energy provision and evaluating the adequacy of support [11]. In this case, the patient's baseline BMI exceeded 30, and the combined effects of whole-body edema due to SIRS and fluid resuscitation caused significant weight fluctuations. Using actual body weight in predictive equations increased the overestimation of energy needs, whereas ideal body weight resulted in underestimation [12]. Furthermore, hypocaloric high-protein feeding is often beneficial for critically ill obese patients; however, the hypermetabolic state associated with severe burns raises concerns regarding the safety of restricting caloric intake [2]. In addition, signs of malnutrition, such as muscle wasting, are harder to detect in patients with burns, necessitating meticulous evaluation. During the 200-day hospital stay, only 2 NST consultations were conducted. Although nutritional screening and dietary interventions are provided by clinical dietitians, their frequencies are limited.

In conclusion, this case emphasizes the importance of active NST involvement and continuous monitoring in evaluating and adjusting nutritional support to manage severe burns effectively. Insufficient intake caused by systemic deterioration and frequent complications led to nutritional decline, necessitating early NST interventions and tailored nutritional strategies to address these challenges. Nutritional interventions that reflect patient preferences and dietary habits can help increase intake. When food intake is insufficient, a multidisciplinary team approach should be used to effectively supplement nutritional deficiencies. Sharing nutritional goals and strategies with patients further enhances patient outcomes. This case highlights the importance of multidisciplinary collaboration, proactive nutritional interventions, and continuous follow-up to optimize treatment for patients with severe burns.

REFERENCES

1. Cho YS. Nutrition therapy in major burns. *J Clin Nutr* 2014;6:45-50. [CROSSREF](#)
2. Carson JS, Khosrozadeh H, Norbury WB, Herndon DN. Nutritional needs and support for the burned patient. In: Herndon DN, editor. *Total burn care*. 5th ed. Amsterdam: Elsevier; 2018. p. 287-300.e2.
3. Rousseau AF, Losser MR, Ichai C, Berger MM. ESPEN endorsed recommendations: nutritional therapy in major burns. *Clin Nutr* 2013;32:497-502. [PUBMED](#) | [CROSSREF](#)
4. Korean Burn Society. *Total burn care*. 1st ed. Paju: Gunja Publishing; 2021.
5. Wang Y, Jiang J, Liu M, Liu H, Shen T, et al. Estimates of resting energy expenditure using predictive equations in adults with severe burns: a systematic review and meta-analysis. *JPEN J Parenter Enteral Nutr* 2024;48:267-74. [PUBMED](#) | [CROSSREF](#)
6. Dickerson RN, Gervasio JM, Riley ML, Murrell JE, Hickerson WL, et al. Accuracy of predictive methods to estimate resting energy expenditure of thermally-injured patients. *JPEN J Parenter Enteral Nutr* 2002;26:17-29. [PUBMED](#) | [CROSSREF](#)
7. Goutos I. Nutritional care of the obese adult burn patient: a U.K. survey and literature review. *J Burn Care Res* 2014;35:199-211. [PUBMED](#) | [CROSSREF](#)
8. Korean Society of Critical Care Medicine. *Guidelines for nutrition support in the adult critically ill patient*. Seoul: Jin Gihuek Publishing; 2013.
9. Park JY, Mok HS, Sin HS, Noh SY, Kim AS, et al. Nutritional support in massive burn patients. *J Korean Burn Soc* 2003;6:118-23.
10. Korean Dietetic Association. *Manual of medical nutrition therapy*. 4th ed. Seoul: Korean Dietetic Association; 2022.
11. Korean Society of Critical Care Medicine. *Critical care medicine*. 3rd ed. Paju: Gunja Publishing; 2017.
12. Nunez JH, Clark AT. Burn patient metabolism and nutrition. *Phys Med Rehabil Clin N Am* 2023;34:717-31. [PUBMED](#) | [CROSSREF](#)